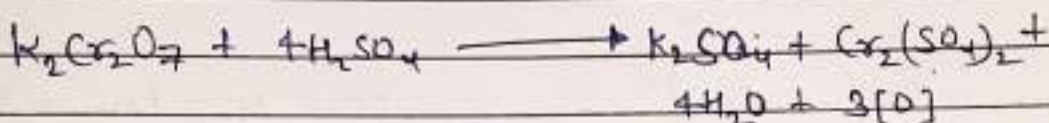


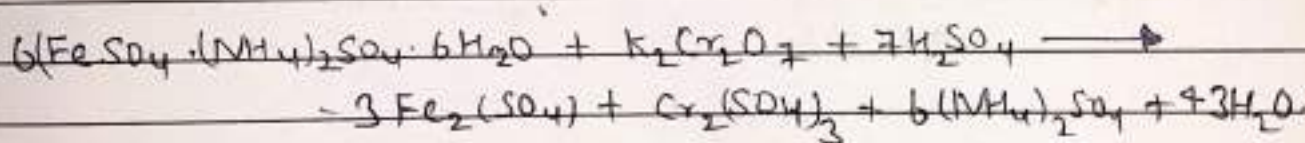
Object :- Determine the strength of unknown Ferrous ammonium sulphate (FAS) solution by titrating it with potassium dichromate solution using N-phenyl anthranilic acid as an internal indicator.

Apparatus / Reagents required :- Burette, pipette, conical flask (100ml), known & unknown FAS solution, unknown $K_2Cr_2O_7$ solution, N-phenyl anthranilic acid, sulphuric Acid (10%) etc.

Theory :- In acidic medium (HCl or H_2SO_4) $K_2Cr_2O_7$ solution acts as an oxidizing agent & gives three atoms oxygen for oxidation of reducing agent.



The oxygen so liberated oxidize Fe^{+2} to Fe^{+3} in FAS solution by adding 5-6 drops N-phenyl anthranilic acid as an internal indicator, the green colour of solution changes purple reddish tinge at the end point.



Procedure :- (1) Prepare standard solution FAS in distilled water.

(2) Fill the burette with known $K_2Cr_2O_7$ solution.

(3) Take 10 ml of standard (FAS) solution & 10 ml Sulphuric

Teacher's Signature _____

Calculation

For standard FAS solution with $K_2Cr_2O_7$

$$N_1 V_1 = N_2 V_2$$

$$N_1 \times 9.6 = N_2 \times 10$$

$$N_1 = \frac{N_2}{3 \times 9.6}$$

here N_2 = Normality of known solution = $N_{K_2Cr_2O_7}$
 V_2 = 10 ml
 V_1 = volume of $K_2Cr_2O_7$
 N_1 = Normality of $K_2Cr_2O_7$

Now for standard unknown FAS solution with same $K_2Cr_2O_7$

$$N_3 V_3 = N_4 V_4$$

$$\frac{N}{3 \times 9.6} \times 5.4 = N_4 \times 10$$

$$N_4 = \frac{N \times 5.4}{30 \times 9.6}$$

$$N_4 = 0.01875 N$$

here N_3 = Normality of $K_2Cr_2O_7$ = N_1

N_4 = Normality of unknown FAS

V_4 = Volume of unknown FAS

V_3 = Volume of known $K_2Cr_2O_7$

acid in a conical flask.

4. Add 5-6 drops of N phenyl anthranilic acid into the conical flask.
5. Add $K_2Cr_2O_7$ solution from burette into (FAS) solution with constant shaking to observe the colour change.
6. Add solution drop wise until the solution in conical flask green due to the formation of $Cr_2(SO_4)_3$.
7. Addition of $K_2Cr_2O_7$ solution suddenly changes the green colour to purple reddish tinge. This indicates the end point.
8. Repeat the titration till concordant reading is obtained.
9. Follow the titration procedure for the unknown solution to obtain the burette reading.

Observation:-

(A) Titration between standard FAS solution & Indeterminate $K_2Cr_2O_7$.

Sl. No.	Volume of known FAS solution (ml)	Volume of $K_2Cr_2O_7$		Difference (ml)	Concordant Reading
		Initial	Final		
1.	10	0	9.6	9.6	9.6
2.	10	9.6	19.2	9.6	9.6

Now the strength of unknown FAS

$$\begin{aligned}\text{Strength} &= N_1 \times \text{eq. weight} \\ &= 0.01875 \times 332.16 \\ &= 7.353 \text{ gm/Lt}\end{aligned}$$

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18) Titration between unknown FAS solution & $K_2Cr_2O_7$

Sl. No.	Volume of unknown FAS solution (ml)	Volume of $K_2Cr_2O_7$		Difference (ml)	Concordant Reading
		Initial (ml)	Final (ml)		
1.	10	0	5.5	5.5	5.4
2.	10	5.5	10.9	5.4	

Result:- The strength of unknown FAS solution is 7.353 gm/Lt.

Precautions:-

- i) All the apparatus should be washed properly.
- ii) $K_2Cr_2O_7$ solution should be added drop wise.
- iii) End point should be observed carefully.

Industrial Application:-

- i) It is used to determine the strength of unknown solution.
- ii) Titration is used to analyse acid rain.
- iii) Titration are used to determine the concentration of contaminants in surface water.
- iv) It is also used in food & beverage industry to analyse the standard concentration of specific additives.
- v) FAS is used in the factories dose meter to measure high doses of gamma rays.

* Viva Questions:-

Q1 Why dil. H_2SO_4 is added in the preparation by FAS solution?

Ans Dilute Sulphuric Acid (H_2SO_4) is added to prevent the hydrolysis of ferrous sulphate. It prevent of the Fe^{+2} ion to Fe^{+3} ions (Yellow)

Q2 What do you mean by volumetric analysis?

Ans Volumetric analysis any method of quantitative, chemical analysis in which the amount of a substance is determined by measuring to volume that is occupied.

Q3 What do you understand by concordant reading?

Ans Readings have been taken several times & the readings are identical or close to each other. This is known as concordant reading.

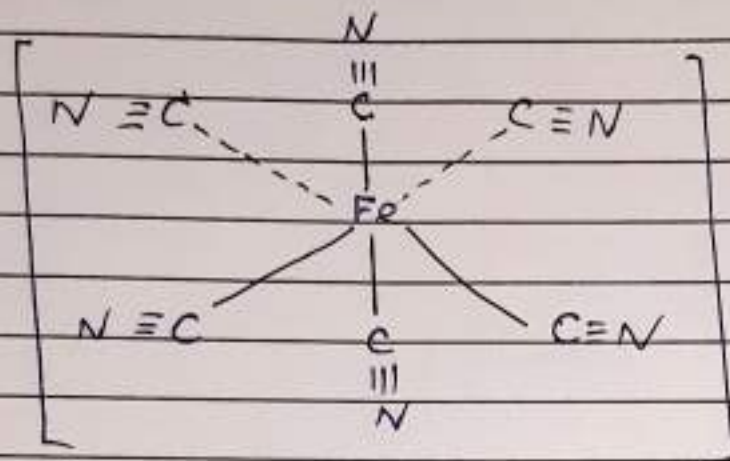
Q4 Explain the types of indicator.

Ans Indicator are the substance that change colour when they are added to acidic or alkaline solutions.

Litmus, phenolphthalein and methyl orange, N-phenyl anthranilic acid and potassium ferricyanide, are used in the laboratory change in property is observed when they come in contact with an acidic or basic solution.

Q5 Which indicator is used in this s. titration. write its structure.

Ans Potassium ferricyanide is used in this titration as external indicator.



Structure

Q6 Explain the principle involved in this titration.

Ans Ferrus ammonium sulphate is a stable double salt with Fe²⁺ being its active constituent. It is a compound of iron and sulphate in which the ratio of ammonium iron (Fe²⁺) to sulphate ions (SO₄²⁻) is 2:1:2.

Q7 Why distilled water is used to prepare standard solution?
Because it does not contain impurities like dissolved iron ions.